



Hepatic Injury, Repair and Fibrosis

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Gastrointestinal block



Acetaminophen hepatotoxicity

Acetaminophen overdose is the most common cause of acute liver failure.

Glutathione depletion leads to accumulation of reactive metabolites.

Increased NAPQI formation causes decreased mitochondrial respiration.



Patterns of hepatocyte necrosis

Centrilobular necrosis is the first manifestation of acetaminophen toxicity.
Bridging necrosis is characterized by bands of necrosis linking portal tracts.

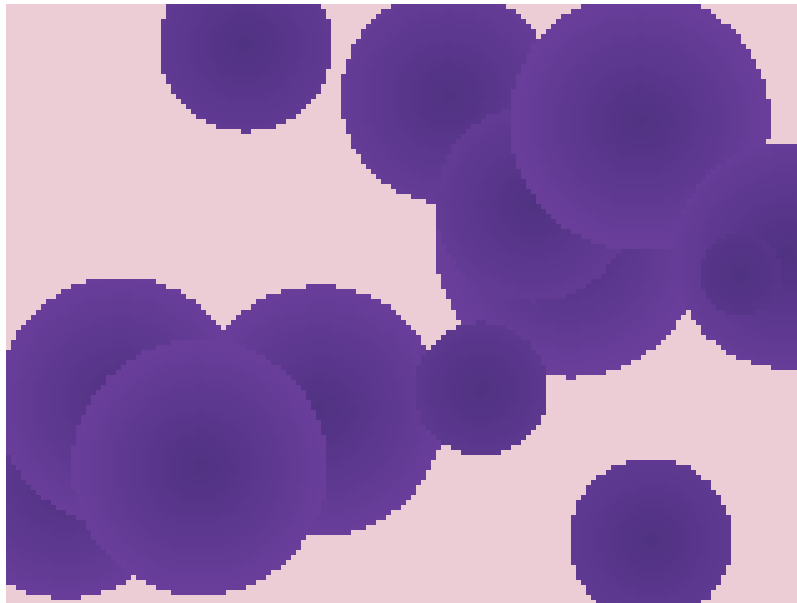


Figure 3.1: centrilobular necrosis with preserved periportal hepatocytes



Alcoholic liver disease

Chronic alcohol use results in macrovesicular steatosis.

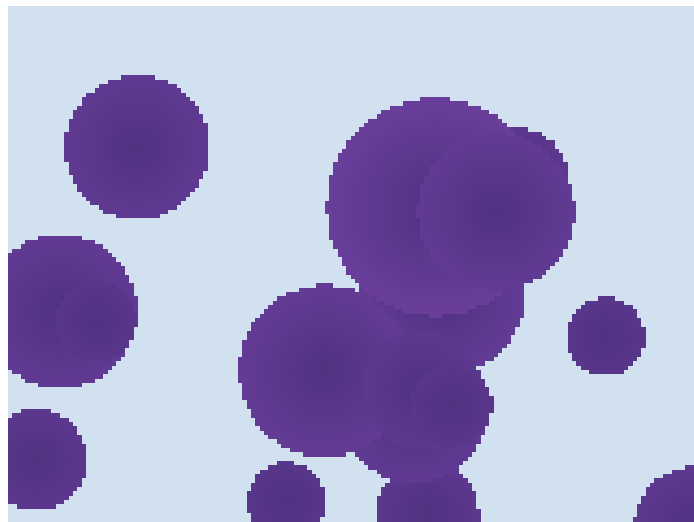
Mallory bodies are caused by aggregation of damaged intermediate filaments.

Alcoholic hepatitis is defined as hepatocyte ballooning with neutrophilic infiltration.



Biochemical markers

Alanine aminotransferase is the most specific marker of hepatocellular injury.
Serum aminotransferases rise within twelve hours of the toxic insult.





Regeneration and fibrosis

Hepatocyte replication restores liver mass by two weeks.

Portal fibrosis follows repeated episodes of hepatocyte injury in many patients.

Stellate cell activation -> collagen deposition -> septal fibrosis -> cirrhosis



Viral hepatitis B

Chronic hepatitis B shows hepatocytes with finely granular cytoplasm.

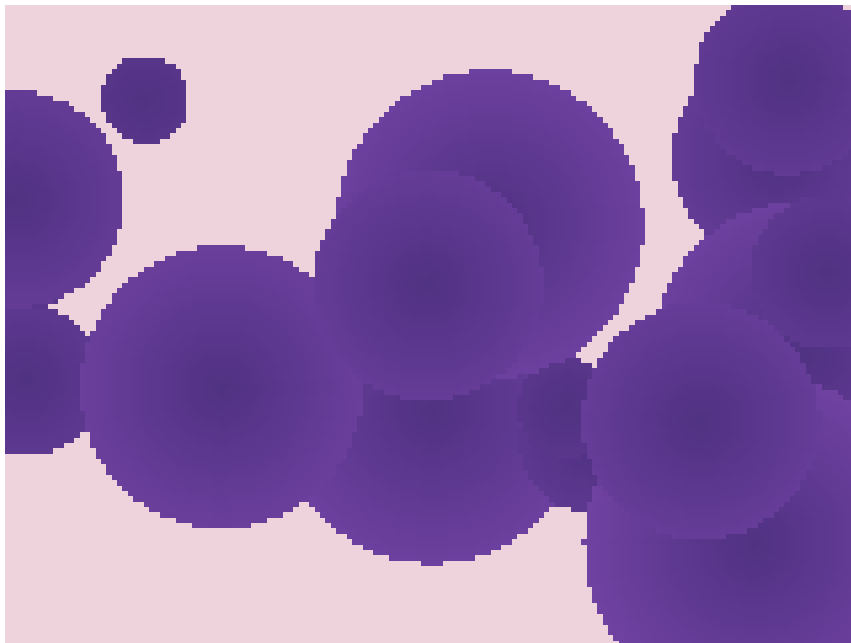


Figure 7.1: ground-glass hepatocytes in chronic hepatitis B



Viral hepatitis A

Acute hepatitis A produces scattered apoptotic hepatocytes.

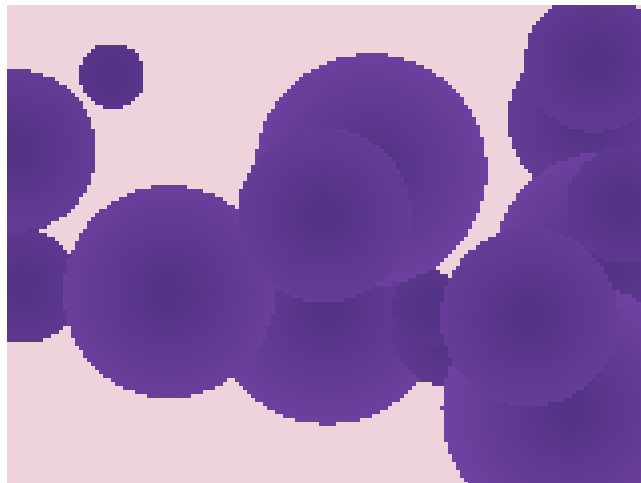


Figure 8.1: councilman bodies in acute hepatitis A



Cholestasis

Conjugated bilirubin accumulates within canaliculi during cholestasis.

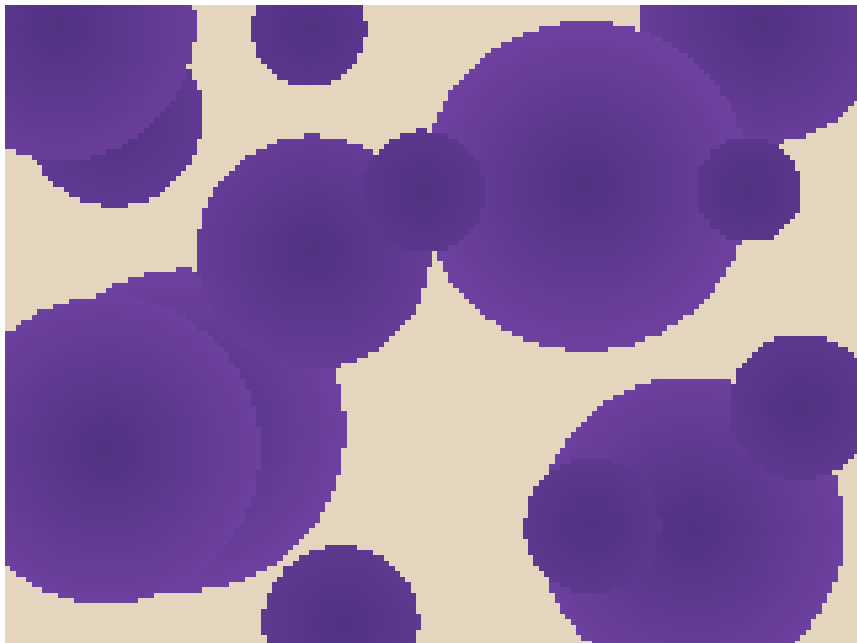


Figure 9.1: canalicular bile plugs



Cholestasis revisited

Prolonged cholestasis damages bile ductules through retained bile acids.



Figure 10.1: canalicular bile plugs



References

Macswheen's Pathology of the Liver, 8th edition.

Thank you